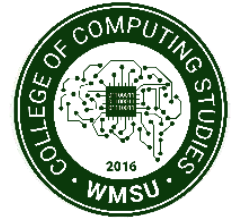




Republic of the Philippines
Western Mindanao State University
College of Computing Studies
DEPARTMENT OF INFORMATION TECHNOLOGY
Zamboanga City



HalalVerify: A Computer Vision System for Real-Time Ingredient Additive Parsing and Halal Logo Authentication

A CAPSTONE PROJECT

Presented to the Faculty of the

College of Computing Studies
Western Mindanao State University

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Information Technology

CUENTO, COLLEN B.
MIRA, KENNETH Q.
VILLANUEVA, ASHLEY NICOLE N.
Researchers

TAHIL, SALIMAR B., MEnggEd
Adviser

June, 2027



WESTERN MINDANAO STATE UNIVERSITY
Zamboanga City, 7000 Philippines

College of Computing Studies
Information Technology Department

CERTIFICATE OF PANEL APPROVAL

The **Capstone Proposal**, attached hereto, entitled "**HalalVerify: A Computer Vision System for Real-Time Ingredient Additive Parsing and Halal Logo Authentication**", prepared and submitted by **COLLEN B. CUENTO, KENNETH Q. MIRA, ASHLEY NICOLE N. VILLANUEVA** in partial fulfillment of the requirements for the degree **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY**, is hereby recommended for approval.

Chairperson

Date

Member

Date

Member

Date

Adviser

Date

This **Capstone Proposal** is approved in partial fulfillment of the requirements for the degree **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY**

MR. JASON A. CATADMAN, MIT
Department Head, IT

Date

ENGR. MARK L. FLORES, PhD
Dean, CCS

Date

EXECUTIVE SUMMARY

In modern retail and food service environments, traditional manual halal verification frequently introduces consumer friction, resulting in purchasing uncertainty, prolonged ingredient review times, and regular authenticity doubts for the Muslim community. Furthermore, existing consumer methods, typically relying on manual dictionary lookups or text-only search engines, are time-consuming, restricted in scope, and prone to inaccurate compliance detection when complex chemical designations or fraudulent logos are evaluated. To mitigate these real-world consumer vulnerabilities, this capstone project presents **HALALVERIFY: A Computer Vision System for Real-Time Ingredient Additive Parsing and Halal Logo Authentication**.

HalalVerify modernizes consumer safety by combining mobile smartphone cameras with deep learning architectures to execute automated visual inspection. Operating via an edge-optimized verification pipeline, the system evaluates captured images to instantly parse textual attributes, localize accredited markings, and classify product compliance into distinct states: HALAL, HARAM, or DOUBTFUL/UNVERIFIED. To ensure strict data privacy, image processing for feature extraction occurs strictly within local volatile memory (RAM); no raw camera frames, video feeds, or personal user identity signatures are ever saved to permanent storage or transmitted over network ports without authorization.

The system delivers comprehensive features, including a Label Scanner and Parser powered by a custom-trained YOLOv8-Nano object detection model for Halal logo localization, alongside EasyOCR for parsing alphanumeric ingredient text and chemical E-numbers. Parsed strings are dynamically cross-referenced against a custom Supabase relational database backend containing categorized additive safety profiles. Furthermore, HalalVerify includes an interactive Establishment Search Index that transforms nationwide retail listings into actionable geolocation data via Leaflet.js, displaying automated compliance standing reports, active registration statuses, and verified inventory manifests. These tools enable consumers, retailers, and regulatory inspectors to optimize product verification and audit workflows based on data-driven deep learning models rather than manual lookup methods.

Development is governed by the Agile methodology, ensuring progressive software iterations and robust system performance. Quantitative system performance will be rigorously benchmarked using standard computer vision evaluation parameters, specifically Precision,

Recall, mean Average Precision (mAP50), and Character Error Rate (CER), measured across model iterations under varying physical lighting and angle variations. User acceptance and overall system utility will be further validated against the ISO/IEC 25010 software quality standards and the System Usability Scale (SUS), ensuring the software meets the operational and cognitive needs of consumer cohorts and regulatory inspectors.

Ultimately, HalalVerify establishes a highly scalable, edge-optimized infrastructure for smart consumer protection in the Philippine retail and food service sectors. The deployment of this system directly supports SDG 9 (Industry, Innovation, and Infrastructure) and SDG 12 (Responsible Consumption and Production) by reducing verification friction, mitigating exposure to non-compliant ingredients through database-driven transparency, and significantly elevating the overall safety experience for consumers, using target cohorts within Zamboanga City as an experimental testing boundary.

DEDICATION

To our beloved families, who provided endless patience, emotional support, and financial assistance throughout our academic journey;

To our friends and classmates, who shared countless sleepless nights and words of encouragement during times of implementation doubt;

And above all, to the Almighty God, for the divine wisdom, absolute strength, and sustaining grace that guided us from the project inception to the successful completion of this work.

- Collen, Kenneth, and Ashley

ACKNOWLEDGEMENT

The success of this capstone project would not have been possible without the guidance, support, and contributions of several individuals. The researchers wish to express their sincere gratitude and deepest appreciation to the following:

To our Capstone Advisers, Prof. Jason A. Catadman, and Prof. John Paul I. Arip, for their invaluable technical expertise, constructive critiques, and dedication in steering the development of HalalVerify toward industry standards;

To the members of the Capstone Defense Panel, for their professional insights, rigorous evaluation, and foundational recommendations that significantly sharpened the research design and system architecture during our defense stages;

To the Dean and Faculty of the College of Computing Studies at Western Mindanao State University, for providing a conducive academic environment, rich technical resources, and institutional paradigms necessary to cultivate our engineering competencies;

To our target consumer test groups and local retail managers within Zamboanga City, who generously offered their operational environments and critical minutes during our system evaluation and User Acceptance Testing (UAT) modules;

And to our loving parents and families, whose unconditional love, sacrifices, financial support, and unwavering faith served as our core motivation to persevere and finish our undergraduate degree.

ABSTRACT

Consumer compliance verification remains a persistent friction point for Muslim communities in globalized retail environments, where complex ingredient chemical designations and unverified logos frequently complicate purchasing choices. This capstone project develops HalalVerify, an intelligent computer vision system engineered as an installable Progressive Web Application (PWA) to automate consumer safety workflows via mobile edge computing. Operating through a multi-stage computer vision pipeline, the platform utilizes a fine-tuned YOLOv8-Nano model to localize certified Halal markings on packaging, cascades the regional coordinates to an EasyOCR engine to parse raw tracking numbers, and cross-references extracted alphanumeric streams and food additives against a custom relational backend database hosted on Supabase. Additionally, the platform incorporates a dedicated nationwide Establishment Search Index, utilizing Leaflet.js and Supabase PostGIS spatial extensions to map verified compliance transparency across local markets, using regional dining hubs within Zamboanga City as an empirical evaluation testbed. To safeguard user privacy, image processing occurs strictly inside local volatile memory layers without retaining facial profiles or raw camera footage files. Quantitative experimental trials will evaluate model efficiency via Precision, Recall, mean Average Precision (mAP50), and Character Error Rate (CER) matrices under varying lighting and structural distortion parameters, while subjective platform usability will be quantified using the standard System Usability Scale (SUS). The resulting framework demonstrates that browser-based, edge-optimized computer vision systems present a fast, scalable, and independent alternative to restricted commercial APIs, significantly driving consumer transparency and digital infrastructure automation across the domestic retail ecosystem.

Keywords: *Computer Vision, YOLOv8-Nano, EasyOCR, Halal Logo Authentication, Ingredient Additive Parsing, Establishment Search, Progressive Web Application (PWA).*

TABLE OF CONTENTS

TITLE	i
APPROVAL SHEET	ii
EXECUTIVE SUMMARY	iii
DEDICATION	v
ACKNOWLEDGEMENT	vi
ABSTRACT.....	vii
LIST OF TABLES	x
LIST OF FIGURES	xi
LIST OF APPENDICES	xii
LIST OF ABBREVIATIONS	xiii

Chapter Page

1 INTRODUCTION	1
1.1 Project Context	1
1.2 Statement of the Problem	3
1.3 Purpose and Description	4
1.4 Objectives	5
1.4.1 Specific Objectives	5
1.5 Scope and Limitation	7
1.6 Significance of Study	9
1.7 Definition of Terms	10
2 REVIEW OF RELATED LITERATURE AND SYSTEMS	14
2.1 Introduction	14
2.1.1 Vision-Based Systems for Food Product Identification	14
2.1.2 Object Localization on Packaging Interfaces	15
2.1.3 Alphanumeric Character Tracking and Optical Label Extraction	15
2.2 Summary	16
2.3 Synthesis	17
3 TECHNICAL BACKGROUND	18
3.1 Technology Requirements	24

3.1.1 Software Requirements	24
3.1.2 Hardware Requirements	25
3.1.3 Dataset Bounding, Authority Framework, and Validation Split Protocols	26
3.1.4 Volatile Memory Architecture and Network Data-Flow Protocol	27
3.1.5 System Functional Specifications and Payload Architecture	27
3.2 Research Methodology Framework	29
3.2.1 Research Design	29
3.2.2 Respondents and Sampling Technique	29
3.2.3 Research Instruments	29
3.2.4 Data Analysis Plan and Mathematical Formulations	29
REFERENCES	30
APPENDICES	34
CURRICULUM VITAE	35

LIST OF TABLES

Table		Page
1.1	Problem, Objective, and Empirical Evaluation Alignment Matrix	06
1.2	Definition of Terms	10
2.1	Review of Related Works and System Comparison Table	16

LIST OF FIGURES

Figure	Page
3.1 Administrative Local Registry Management	21
3.2 Customer Scanner View – Web Version	21
3.3 Customer Scanner View – Mobile View	22
3.4 Conceptual Framework of the System	23

LIST OF APPENDIXES

Appendix	Page
Appendix A: Data Gathering	53
Appendix B: Testing	54
Appendix C: Evaluation Form	55
Appendix D: Gantt Chart	60
Appendix E: Use Case	61
Appendix F: Plagiarism Certificate	64

CHAPTER 1

INTRODUCTION

1.1 Project Context

The increasing availability of imported and locally manufactured food products has provided consumers with a wider variety of choices in supermarkets, convenience stores, and restaurants. However, for Muslim consumers, ensuring that food products comply with halal requirements remains a significant challenge. Although many products display halal certification logos, consumers often find it difficult to verify whether these certifications are authentic, recognized, or issued by accredited halal certifying bodies such as the National Commission on Muslim Filipinos (NCMF), the Islamic Da'wah Council of the Philippines (IDCP), the Halal Development Institute of the Philippines (HDIP), or the Halal International Chamber of Commerce and Industries of the Philippines (HICCIP). Furthermore, ingredient lists frequently contain scientific names, complex chemical designations, or E-numbers that are unfamiliar to ordinary consumers, making it difficult to determine whether a product complies with halal dietary regulations. These challenges create uncertainty and reduce consumer confidence when purchasing packaged food products.

Existing halal verification methods primarily depend on manual inspection of halal logos, barcode scanning applications, QR code verification, or searching through online databases. While these approaches provide useful information, they often require internet connectivity, manual searching, or access to official registries. Additionally, counterfeit or improperly displayed halal logos cannot easily be identified by consumers through traditional verification methods. As food products become increasingly diverse and globalized, there is a growing need for a faster, more intelligent, and user-friendly verification system that can assist consumers in making informed purchasing decisions.

This verification friction is critically pronounced across the retail and food service sectors of the Philippines, where consumers must navigate a highly diverse marketplace of local goods, domestic regional items, and international imports. This challenge is highly visible within regional trading hubs like Zamboanga City, which serves as an optimal domestic testing boundary due to its unique cultural demographics and flood of products

from neighboring economic regions. Because the Philippines lacks a single, consolidated public API or open-access digital infrastructure linking all the accredited Halal Certification Bodies (HCBs) such as the IDCP, HDIP, and regional entities, ordinary consumers frequently encounter packaging with unverified certification stamps or obscure chemical E-number designations. Consequently, the lack of data-driven transparency within the domestic retail supply chain drives purchasing uncertainty, emphasizing a clear nationwide demand for an independent mobile edge-verification Progressive Web Application (PWA) whose technical deployment will be experimentally evaluated and field-tested within the local retail environments of Zamboanga City.

Furthermore, a significant operational gap exists regarding structural lookup accessibility for local food businesses. Consumers inside a specific municipality often lack public methods to check if an establishment maintains an active compliance certification or hosts specific certified products. Integrating spatial queries with label analysis addresses this transparency gap.

Recent advancements in Artificial Intelligence (AI), Computer Vision, Optical Character Recognition (OCR), and Deep Learning have enabled the development of intelligent image-based verification systems capable of recognizing objects, extracting textual information, and validating compliance data. Object detection models such as YOLOv8-Nano provide accurate real-time detection of certification logos, while OCR engines such as EasyOCR enable automated extraction of ingredient lists and E-numbers from captured images. These technologies provide an opportunity to automate halal verification without requiring specialized hardware or extensive manual intervention.

To address these challenges, the proponents propose HalalVerify: A Computer Vision System for Real-Time Ingredient Additive Parsing and Halal Logo Authentication. The proposed system is a Progressive Web Application (PWA) that utilizes a multi-stage computer vision structure. The first module detects accredited halal certification logos on product packaging using YOLOv8-Nano. The second module extracts raw ingredient text using EasyOCR and cross-references parsed food additives against a custom Supabase relational database backend containing categorized compliance profiles. Finally, the system introduces a dedicated establishment query index to verify the legal status of local food hubs, providing consumers, retailers, and regulatory personnel with a fast, accessible, and

intelligent tool for halal verification that promotes consumer confidence, food safety, and regulatory compliance.

1.2 Statement of the Problem

The increasing number of packaged food products available in retail stores has made halal verification more challenging for Muslim consumers. While many products display halal certification logos, consumers often have difficulty determining whether these certifications are genuine or issued by recognized halal certifying organizations. Additionally, ingredient labels frequently contain complex chemical names and food additives that are unfamiliar to consumers, making it difficult to identify ingredients that may be prohibited under halal dietary regulations. Existing verification methods largely rely on manual inspection, barcode scanning, or online database searches, which are often time-consuming. Furthermore, consumers lack an interactive mechanism to actively search for local dining spots or retail establishments to cross-reference whether they hold valid compliance rankings or sell authenticated goods.

This study aims to develop HalalVerify: A Computer Vision System for Real-Time Ingredient Additive Parsing and Halal Logo Authentication, which utilizes computer vision, deep learning, and OCR technologies to automatically parse ingredient labels, authenticate halal logos, and lookup regional food establishment criteria through a smartphone camera.

Specifically, this study seeks to answer the following questions:

1. How can a computer vision pipeline using YOLOv8-Nano detect and localize accredited halal certificate logos on packaged food products?
2. How can EasyOCR parse scientific chemical additive strings and E-numbers from ingredient labels to cross-reference them against a custom relational backend database on Supabase?
3. How can a relational database index be structured within a PWA to allow consumers to query local establishments regarding their halal compliance status and authenticated product availability?
4. What is the performance of the proposed HalalVerify system in terms of Precision, Recall, mean Average Precision (mAP_{50}), and Character Error Rate (CER)?

5. How acceptable is the HalalVerify system to consumers and halal inspectors based on the System Usability Scale (SUS)?
6. How reliable is the proposed system under varying image conditions such as different lighting environments, camera angles, blurred images, and partially obstructed labels?

1.3 Purpose and Description

Muslim consumers frequently encounter uncertainty when purchasing packaged food products because verifying halal compliance often requires manual inspection of certification logos, careful examination of ingredient lists, or searching through online databases. This process becomes more difficult when ingredient labels contain scientific names or food additives that are unfamiliar to consumers. Moreover, counterfeit or unverified halal logos can further reduce consumer confidence and increase the risk of purchasing products that do not comply with halal standards.

Current halal verification methods generally rely on barcode scanners, QR code verification systems, or manual database searches. Although these methods provide useful information, they require additional user effort and cannot automatically parse chemical additive text directly from images. These limitations emphasize the need for an intelligent and automated verification system that combines computer vision with database cross-referencing.

To address these challenges, the proponents propose HalalVerify: A Computer Vision System for Real-Time Ingredient Additive Parsing and Halal Logo Authentication. The system utilizes YOLOv8-Nano to detect accredited halal certification logos and EasyOCR to extract alphanumeric ingredient text and E-numbers from product packaging. Parsed text is dynamically evaluated against a custom relational backend database maintained on Supabase. Additionally, it integrates an administrative search asset enabling consumers to dynamically look up local food spots and track their current certified parameters. Verification results are presented through a user-friendly Progressive Web Application that classifies products as Halal, Haram, or Doubtful/Unverified. By automating the verification process, HalalVerify provides consumers with faster and more reliable access to halal information while supporting retailers and regulatory authorities in promoting food safety and compliance.

1.4 Objectives

The primary objective of this study is to design and develop HalalVerify: A Computer Vision System for Real-Time Ingredient Additive Parsing and Halal Logo Authentication. The proposed system aims to assist consumers, retailers, and halal inspectors by providing an intelligent and automated solution for verifying halal product labels, authenticating logos, and checking establishment standings using computer vision and OCR technologies.

1.4.1 Specific Objectives

- To develop a real-time object detection module using YOLOv8-Nano that identifies and localizes accredited halal certification logos on packaged food products.
- To develop an OCR-based module using EasyOCR that extracts raw chemical ingredient test strings and E-numbers from product packaging.
- To design and integrate a custom relational backend database on Supabase to cross-reference parsed ingredients and logo credentials against categorized compliance profiles.
- To implement a search query functionality within the backend that allows users to lookup specific retail and dining establishments to verify their overall halal status and associated product listings.
- To evaluate the technical performance of the machine learning engines using Precision, Recall, mean Average Precision (mAP_{50}), and Character Error Rate (CER) benchmarks.
- To assess the usability and overall acceptability of the HalalVerify PWA using the System Usability Scale (SUS).
- To ensure rigorous methodological consistency, the research components, core inquiries, and empirical evaluation success thresholds of this study are systematically mapped in Table 1.1.

Table 1.1: Problem, Objective, and Empirical Evaluation Alignment Matrix

System Evaluation Component	Linked Research Question	Specific Project Objective	Primary Evaluation Data Source	Target Performance Metric	Empirical Success Criterion
Halal Logo Scan and Identification	RQ1: Visual localization and bounding-box reliability under varying image inputs.	Objective 1: To develop a real-time object detection module using YOLOv8-Nano.	Custom repository of 400 annotated packaging images collected from stores.	Precision, Recall, mAP50	Mean Average Precision (mAP50) $\geq$ 85% under standard lighting.
Ingredient String Parsing	RQ2: Accuracy of parsing scientific chemical identifiers from food wrappers.	Objective 2: To develop an OCR-based module using EasyOCR to parse additives.	Cropped image matrix blocks containing text packaging labels.	Character Error Rate (CER)	Character Error Rate (CER) $\leq$ 10% on clear, legible labels.
Establishment Search Engine	RQ3: Database indexing performance within the browser-based PWA framework.	Objective 3: To implement a search query function for nationwide establishments.	Curated database registry of Philippine retail and dining food hubs.	Query Latency / Response Time	End-to-end data retrieval response time $\leq$ 2.0 seconds.

Environmental Robustness Testing	RQ6: Multi-stage vision pipeline resilience across physical degradation states.	Objective 4: To evaluate system metrics under varying stress environments.	Stress-test image dataset variations (varying angles, blur, low-light).	Comparative Metric Delta	Total accuracy degradation $\leq 15\%$ under mild image distortion.
User Acceptance Testing (UAT)	RQ5: Subjective end-user usability and accessibility perceptions.	Objective 5: To assess platform usability using the standard SUS instrument.	Purposive sample size of 40 local evaluators in Zamboanga City.	System Usability Scale (SUS) Score	Composite average SUS valuation score ≥ 68.0 (Above Average).

1.5 Scope and Limitation

This study focuses on the design, development, and evaluation of HalalVerify: A Computer Vision System for Real-Time Ingredient Additive Parsing and Halal Logo Authentication. The proposed system is an installable Progressive Web Application (PWA) optimized for standard mobile smartphones. The system provides three primary operational modules: the Ingredient Additive Parser, the Halal Logo Authenticator, and the Establishment Directory Search. The Ingredient Additive Parser allows users to scan or upload an image of a packaging label, after which EasyOCR extracts raw text strings and E-numbers to cross-reference them against chemical additive safety records in a custom Supabase relational database. Based on the parsed results, the product is classified as Halal, Haram, or Doubtful/Unverified. The Halal Logo Authenticator utilizes a fine-tuned YOLOv8-Nano object detection model to isolate and identify recognized Halal certification marks on product packaging, cross-referencing visual logo features against accredited certifying bodies stored in the Supabase backend registry. Finally, the Establishment Directory Search features a text and category-driven lookup tool allowing users to query

local dining and retail establishments within Zamboanga City to verify their active compliance standing or view authenticated product listings.

The primary target users of the system are Muslim consumers, food retailers, restaurant owners, and halal regulatory inspectors. The technical stack utilizes YOLOv8-Nano, EasyOCR, and OpenCV for computer vision processing; Python with FastAPI or Flask for backend services; React.js and Tailwind CSS for the Progressive Web Application frontend; and Supabase (PostgreSQL) for database management. Technical benchmark testing will be conducted using a dataset of approximately 300–500 packaged grocery product label images collected from select retail stores within Zamboanga City.

However, the scope of this study is bound by several specific technical and operational limitations. Regarding database integration and external registry disconnection, the system operates strictly as an independent decision-support tool and does not connect via live APIs to external third-party databases, government registries, or official certifying bodies, such as the Islamic Da'wah Council of the Philippines (IDCP). All logo credentials and additive classification rules are cross-referenced exclusively against a custom-curated relational database hosted on Supabase. Consequently, while the system's logo detection module utilizes YOLOv8-Nano to identify and localize visual features of recognized Halal marks, the Halal Logo Authenticator module will not operate as an isolated pixel-matching classifier, thereby resolving the loophole where computer vision alone cannot verify a physical photocopy or fraudulent logo. Instead, visual authentication will be achieved via a multi-stage software architecture. The fine-tuned YOLOv8-Nano model will handle localized graphic discovery, cropping the region of interest (ROI) containing the certification seal. Once isolated, the client-side engine will cascade the frame segment to an EasyOCR character extraction model to read unique alphanumeric certificate or license tracking number printed on the package. This extracted string will be dynamically queried via secure network connections against an administrative reference database hosted on Supabase (PostgreSQL), cross-referencing records linked to active Halal Certification Bodies (HCBs). Therefore, authentication is programmatically defined by the successful correlation of visual attributes with valid database registry parameters.

Furthermore, the system's performance is sensitive to environmental and image capture conditions. Detection and OCR extraction accuracy depend on lighting, camera angles, focus, and resolution, meaning system performance may degrade when processing wrinkled, damaged, partially obstructed, or poorly lit packaging labels. All generated system outputs are advisory and informational only, serving as decision-support tools rather than legal substitutes for formal verification by accredited halal certifying bodies. Additionally, the study explicitly excludes the verification of raw or fresh meat products, kitchen cross-contamination tracking, and restaurant food preparation methods. The system also relies on active internet connectivity for real-time database lookups unless a cached local dictionary has been successfully bundled within the browser service worker. Finally, due to strict project execution timelines, usability testing using the System Usability Scale (SUS) will involve a targeted respondent group of approximately 30 to 50 participants.

1.6 Significance of Study

The development of HalalVerify introduces an automated computer vision approach to consumer protection. By automating the detection of halal certification logos and extracting ingredient additive information using only a smartphone camera, the system reduces the need for manual dictionary searching and enables consumers to make faster, more confident purchasing decisions.

The proposed system provides significant benefits to Muslim consumers by simplifying the process of verifying halal food products. Instead of manually interpreting complex scientific ingredient labels or researching unfamiliar food additives (E-numbers), users obtain structured verification results within seconds.

For food retailers and restaurant owners, the system serves as an additional verification mechanism to ensure that products offered to customers align with halal requirements, enhancing business transparency and strengthening consumer trust.

The study is also beneficial to halal regulatory agencies and inspectors by providing a portable decision-support tool during preliminary field audits. Although the system does

not replace official certification procedures, it offers a rapid method for identifying non-compliant or doubtful ingredient profiles that may require further investigation.

From an academic perspective, the study contributes to the research domain of applied computer vision, optical character recognition, and mobile consumer protection. It demonstrates how deep learning models such as YOLOv8-Nano and EasyOCR can be seamlessly integrated with a custom database engine to address real-world food transparency challenges. The project directly aligns with Sustainable Development Goal (SDG) 9: Industry, Innovation, and Infrastructure and SDG 12: Responsible Consumption and Production.

1.7 Definition of Terms

Defined terms are those whose meanings in the study are particular or unique. Give a minimum of fifteen. The researcher must credit his sources for any definitions he uses, even if they are from dictionaries, books, journals, newspaper articles, and other sources. Technical and conceptual definitions are those that are derived from published sources.

Table 1.2: Definition of Terms

Term	Definition
1. Artificial Intelligence (AI)	Artificial Intelligence refers to the capability of computer systems to perform tasks that normally require human intelligence, such as learning, reasoning, and decision-making. (Russell & Norvig, 2021)
2. Computer Vision	Computer Vision is a field of Artificial Intelligence that enables computers to acquire, process, and interpret visual information from digital images or videos. (Szeliski, 2022)
3. YOLOv8-Nano	An ultra-lightweight, edge-optimized real-time object detection model developed by Ultralytics that identifies and classifies objects within localized image and video streams. (Jocher et al., 2023)

Term	Definition
4. Object Detection	Object Detection is a computer vision technique that identifies and locates objects within images or video frames. (Redmon & Farhadi, 2018)
5. Optical Character Recognition (OCR)	A technology that converts printed or handwritten text contained within visual inputs into machine-readable digital text. (Gudi, 2025)
6. EasyOCR	An open-source OCR library that supports multilingual text recognition using deep learning models for extracting text from images. (Jaided, 2024)
7. Custom Backend Database	A relational cloud database built on Supabase (PostgreSQL) that stores structured records of chemical additives, accredited logos, and establishment profiles for instant cross-referencing (Supabase, 2024).
8. Ingredient Additive Parsing	The automated computer-vision process of scanning, segmenting, and isolating text strings representing food additives or E-numbers to evaluate their compliance status (Haseeb, 2025).
9. Halal Certification	Official recognition issued by an authorized halal certification body confirming that a product or service complies with Islamic dietary laws and halal standards. (SMIIC, 2019)
10. Halal Logo	An officially approved symbol placed on products or establishments indicating compliance with halal certification requirements established by an accredited halal certifying body. (SMIIC, 2019)
11. Progressive Web Application (PWA)	A web-delivered software application built using standard web tech (HTML, JS, React) that functions seamlessly as an installable, cross-platform mobile application (Biørn-Hansen et al., 2020).
12. Web-Based System	A web-based system is an application that can be accessed through a web browser over a network. (Pressman & Maxim, 2020)

Term	Definition
13. OpenCV	OpenCV (Open Source Computer Vision Library) is an open-source software library that includes hundreds of computer vision algorithms used for image processing, video analysis, and real-time object detection. (Bradski, G., & Kaehler, A., 2017)
14. Precision	The proportion of correctly identified positive results among all predicted positive results in a model. (Chicco, D., & Jurman, G., 2020)
15. System Usability Scale (SUS)	The System Usability Scale (SUS) is a reliable, industry-standard, ten-item questionnaire used to provide a subjective assessment of the usability of a wide range of products and services, including software applications and web-based systems. It allows researchers to quantify user perceptions of a system's ease of use and learnability. (Lewis, J.R., 2018)
16. ISO/EIC 25010	ISO/EIC 25010 is an international standard that provides a quality model for software systems, establishing criteria to evaluate functional suitability, performance efficiency, compatibility, and usability. (ISO, 2011)
17. Halal Logo Authentication	The multi-stage computing process of localizing a visual halal logo via deep learning, extracting its alphanumeric certificate code via OCR, and validating that text payload against a curated reference registry.
18. Character Error Rate (CER)	The mathematical metric used to measure EasyOCR performance by dividing the total number of character edits (substitutions, deletions, and insertions) by the total number of characters in the ground-truth string.
19. Doubtful Additive	Any chemical food ingredient or E-number code whose compliance status exhibits sourcing ambiguity (such as plant vs. animal fat derivation in E471) and cannot be resolved by text extraction alone, prompting a default system warning display.

Term	Definition
20. False Positive (Authentication Error)	An operational error where the system incorrectly classifies an unindexed, expired, or counterfeit label configuration as 'Authenticated' due to database lookup error or character recognition bugs.
21. Ground-Truth	The verified empirical reality of a testing asset (e.g., a product wrapper sample with a manually verified active IDCP code and halal-compliant ingredients) used as the true baseline comparison to calculate system accuracy.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND SYSTEMS

2.1 Introduction

The following review of related literature and systems for the proposed study, HalalVerify: A Computer Vision System for Real-Time Ingredient Additive Parsing and Halal Logo Authentication, provides a comprehensive, structurally clustered analysis of contemporary research and intelligent platforms in the domains of food recognition, machine learning-driven label extraction, optical character tracking, and computerized dietary compliance mapping. This chapter investigates the engineering designs, algorithmic frameworks, and deployment strategies of relevant systems to establish a verified technical baseline for automated consumer protection.

To maintain strict methodological alignment with the core operational features of HalalVerify, the reviewed literature and existing systems are organized under four distinctive, feature-oriented thematic categories:

1. Vision-Based Systems for Food Product Identification
2. Object Localization on Packaging Interfaces
3. Alphanumeric Character Tracking and Optical Label Extraction

The following literature and systems exhibit comparable structures and thematic relevance.

2.1.1 Vision-Based Systems for Food Product Identification

Recent advancements in deep learning have optimized automatic food asset classification by enabling computer vision models to identify grocery metrics directly from digital image frames. As demonstrated by Rokhva et al. in, lightweight architectures can be successfully fine-tuned using transfer learning and extensive data augmentation to achieve a classification accuracy of approximately 92.97%. This baseline proves that structurally streamlined networks can run with high computational efficiency on consumer mobile layers without overloading local processing threads. While [16] confirms the on-device reliability of broad graphic classification, HalalVerify expands this paradigm by

cascading isolated bounding boxes through secondary text-line parser queues to evaluate legal status attributes.

2.1.2 Object Localization on Packaging Interfaces

Real-time object localization frameworks, specifically the You Only Look Once (YOLO) deep learning architecture model family, provide robust performance parameters for isolating trade emblems on cluttered consumer product wrappers. Andreotti et al. in [17] developed an automated data extraction framework utilizing YOLO models to locate and crop critical semantic fields such as expiration stamps and retail metrics from packaging graphics before executing adaptive threshold filtering. Similarly, Syahputra et al. in [19] proved that embedded convolutional networks can successfully localize certified Halal logos under real-world supermarket illumination constraints. HalalVerify synthesizes these techniques by training a fine-tuned YOLOv8-Nano model to locate both the certified Halal mark and its adjacent license digits, serving as the foundational phase of a multi-stage authentication system.

2.1.3 Alphanumeric Character Tracking and Optical Label Extraction

Optical Character Recognition (OCR) acts as the primary computing bridge for converting printed packaging text blocks into machine-readable digital character arrays. Bhatlawande et al. in [18] constructed an intelligent scanner application that combines OCR text line recognition with a structured reference repository to identify nutritional layouts, sort chemical designations, and flag potential allergens for dietary safety compliance. Furthermore, Choi et al. in [20] proved that deploying mobile edge OCR pipelines allows for instant character string extraction on low-connectivity devices without incurring high latency overheads. HalalVerify builds upon these foundational systems by using EasyOCR to explicitly target scientific chemical strings and E-numbers, routing the parsed string tokens to a relational backend to eliminate visual counterfeiting loopholes.

Table 2.1 Review of Related Works and System Comparison Table				
Features	Rokhva et al. (2024)	Andreotti et al. (2025)	Bhatlawande et al. (2024)	Proposed Study (HalalVerify)
Login and Logout				✓
User Account				✓
Dashboard	✓			✓
Real-Time Camera Inference	✓	✓	✓	✓
Halal Logo Detection				✓
Ingredient Text Parsing		✓	✓	✓
Localized E-Number Lookup			✓	✓
Compliance-Level Classification	✓		✓	✓
Document Layout Segmentation				✓
Certificate Validity Flagging		✓		✓
Geolocation Search Index				✓
Local Volatile RAM Processing				✓
User Acceptance Testing				✓

2.2 Summary

The reviewed literature demonstrates that Artificial Intelligence, Computer Vision, YOLO-based object detection, and Optical Character Recognition (OCR) are highly effective technologies for automating food recognition, product label detection, and textual information parsing, [17], [18]. Previous studies have proven that lightweight deep learning models can accurately identify food items, isolate packaging attributes, and extract complex ingredient strings while maintaining computational efficiency on standard mobile devices, [20]. However, existing platforms focus predominantly on general food classification, retail inventory tracking, or basic nutritional and allergen auditing, [18]. They lack a unified consumer protection framework specifically tailored to Halal compliance verification. To

address this research gap, HalalVerify integrates YOLOv8-Nano for Halal logo localization, EasyOCR for ingredient text extraction, a custom Supabase relational database for E-number and additive cross-matching, and a local Establishment Directory Search into a single Progressive Web Application (PWA), [30].

2.3 Synthesis

The reviewed literature confirms that Artificial Intelligence, Computer Vision, Object Detection, and Optical Character Recognition (OCR) have substantially advanced the automation of food recognition, label analysis, and data extraction, [17], [18]. Lightweight deep learning architectures enable efficient visual classification, while YOLO object detection models provide fast, bounding-box localization of packaging features, [19]. Concurrently, OCR frameworks have proven reliable for extracting fine-grained ingredient text and additive codes, enabling consumers to make informed dietary decisions, [20].

Despite these technological advancements, existing systems remain fragmented. Current applications focus either on general food item recognition, automated warehouse inventory parsing, or broad allergen tracking, [17], [18]. None of the reviewed systems offer a comprehensive, edge-optimized solution that integrates real-time Halal logo detection, scientific E-number additive parsing, custom relational database cross-matching, and a localized establishment directory into a single user-centric platform. Furthermore, existing implementations rarely address volatile data handling for privacy assurance or offline-first PWA caching for low-connectivity environments.

Based on these findings, the proponents developed HalalVerify to address the identified research gap. The proposed system integrates YOLOv8-Nano for halal logo detection and EasyOCR for ingredient extraction. By combining these technologies, the system provides a comprehensive, real-time, and user-friendly solution for verifying halal product labels and certificates. Unlike previous studies that focus on individual technologies or specific applications, HalalVerify offers an integrated approach that enhances consumer protection, supports regulatory compliance, and contributes to the advancement of intelligent halal verification systems.

CHAPTER 3

TECHNICAL BACKGROUND

HalalVerify is an installable, Progressive Web Application (PWA) designed to automate consumer safety workflows by providing real-time verification of packaged food product labels and establishment compliance standings. The system captures and analyzes visual product attributes through standard mobile smartphone camera interfaces, processing image matrices using a streamlined machine learning pipeline composed of a fine-tuned YOLOv8-Nano object detection model and an EasyOCR text parsing engine. The resulting verification and compliance data are delivered through a responsive PWA framework and displayed on an interactive dashboard that can be securely accessed by Muslim consumers, food retailers, restaurant owners, and regulatory inspectors.

The platform follows a structured Input-Process-Output (IPO) model embedded within a decentralized three-tier architecture. The Input stage consists of live camera feeds or uploaded images of packaged ingredient labels, along with text-based query inputs targeting nationwide dining and retail establishments. During the Process stage, incoming image matrices are preprocessed using OpenCV before being routed to the specialized machine learning engines to localize recognized Halal logo credentials or parse alphanumeric chemical additive strings and E-numbers. Crucially, all camera frames and visual inference operations occur transiently inside local volatile memory (RAM) to ensure absolute data privacy; no raw camera feeds, packaging scans, or personal user identifiers are saved to permanent storage or transmitted externally. The Output stage materializes as a dynamic advisory status display classifying items into specific user-facing labels: HALAL, HARAM, or DOUBTFUL/UNVERIFIED, alongside an interactive establishment profile lookup directory.

The technical design is divided into three main operational layers: Presentation, Application and Logic, and Database.

Presentation Layer

The presentation layer serves as the direct graphical user interface through which distinct stakeholder groups view and interact with the processed halal compliance states. Built using React.js and Tailwind CSS, this layer translates classification flags and lookup data into an intuitive mobile canvas designed for rapid visual parsing.

- **Muslim Consumers:** Consumers interact with a color-coded status dashboard mounted inside the PWA browser interface. The application utilizes dynamic visual banners: Green signifies a detected and authenticated Halal logo or safe additive (**HALAL**); Yellow flags a **DOUBTFUL/UNVERIFIED** chemical additive (displaying warning notices, e.g., "Contains Doubtful E-Number Additives"); and Red designates a strictly prohibited chemical additive (**HARAM**) or a blacklisted label configuration.
- **Food Retailers and Establishment Owners:** Business operators access a secure management view to oversee their establishment directory listings. This interface provides the capability to update inventory catalogs, maintain verified product lists, and display active compliance standings in the local directory index.
- **Halal Regulatory Inspectors:** Inspectors gain entry to a dedicated administrative dashboard layout that renders historical lookup summaries and audit logs, displaying verification statistics and compliance trends within the domestic retail market to assist in localized field operations and decision support.

Application and Logic Layer

The application and logic layer acts as the primary processing engine of HalalVerify, executing computer vision routines, character extraction pipelines, and classification logic.

- **Computer Vision and Detection Module:** Binds directly to native smartphone camera APIs via standard browser media streams. Incoming image matrices are managed using OpenCV for spatial scaling, noise filtering, and thresholding before being routed to a fine-tuned YOLOv8-Nano network for logo bounding box localization and an EasyOCR processing queue for ingredient character mapping.
- **Compliance Classification Mechanism:** Parses extracted text strings and E-numbers against predefined relational reference tables. The system isolates scientific chemical designations and E-number codes, cross-referencing them against a custom additive

reference dictionary hosted on Supabase to determine the corresponding status level (HALAL, HARAM, or DOUBTFUL/UNVERIFIED).

- **Backend API Framework:** Developed using Python and the FastAPI framework, this module handles asynchronous microservice routing, manages JSON payloads from the PWA client, structures access control for business owners, and processes relational table lookups from the persistent storage node with minimal execution latency.

Database Layer

The database layer acts as the centralized data store responsible for logging the curated additive lexicon, registered establishment profiles, user credentials, and administrative logs.

- **Data Storage:** The system utilizes Supabase, an open-source PostgreSQL-based relational database infrastructure. Supabase securely manages structured relational tables containing permitted and prohibited food additives, E-number classification rules, accredited certifying body records, local retail business profiles, and user authentication tables.
- **Data Retrieval:** Leveraging Supabase's indexed REST API framework, the database layer executes high-speed queries to deliver instant additive and establishment lookup responses to the PWA frontend, eliminating heavy processing on the client device.

User Interactions

Administrators: Securely authenticate to manage the reference tables, manually operate validation workflows, configure system options, and oversee or configure the overarching administrative data logs.

Customers: Access unauthenticated scanner functions to view displayed verification results (HALAL, HARAM, or DOUBTFUL/UNVERIFIED) for scanned food items, along with verified logo authentication statuses on the dashboard, and query map nodes without forced sign-in blocks.

Administrative Management

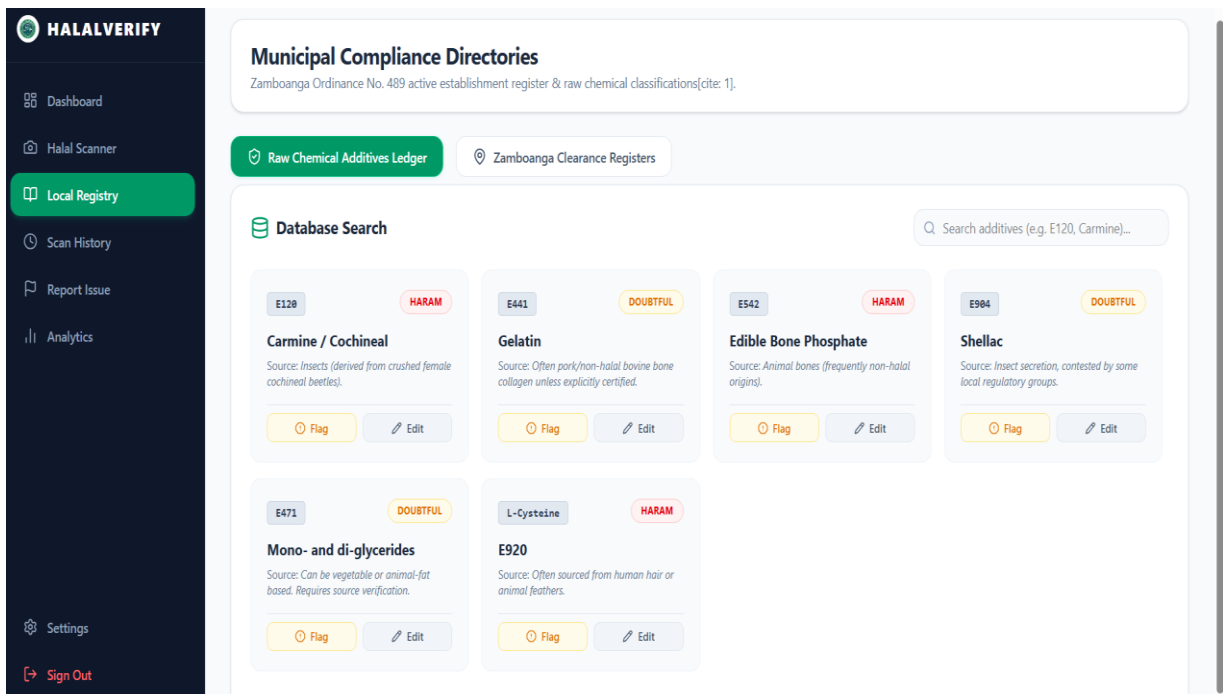


Figure 3.1: Administrative Local Registry Management

Customer Scanner View (Web Version)

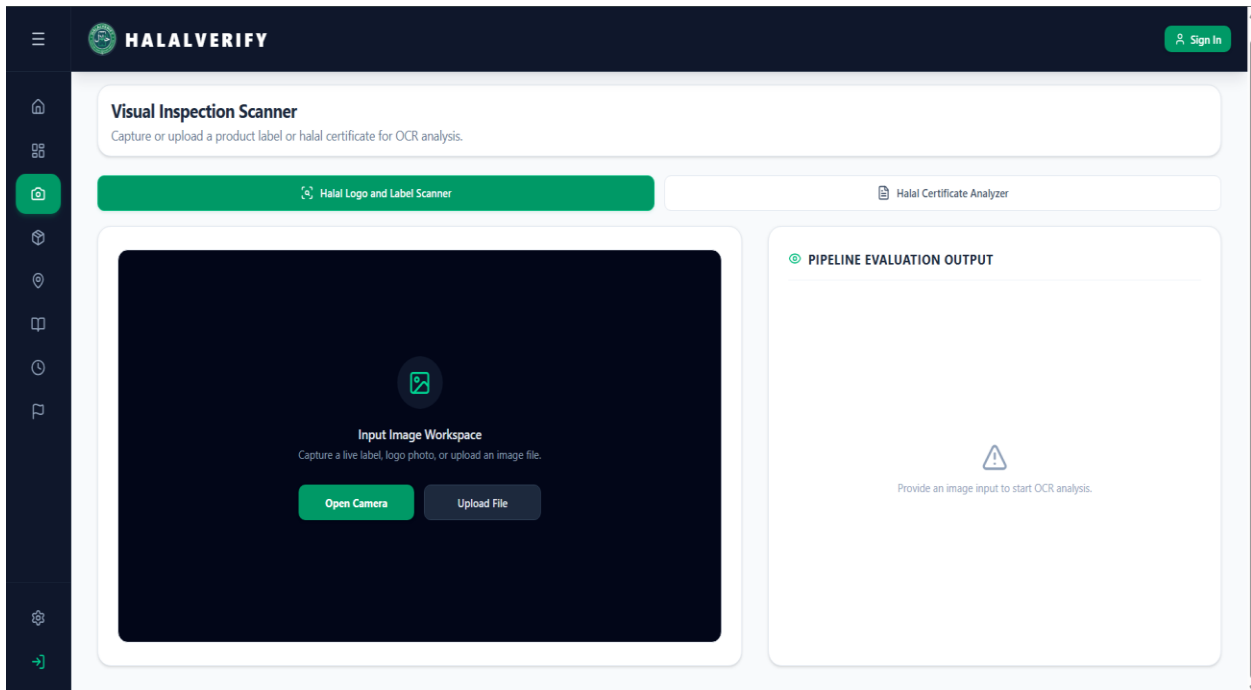


Figure 3.2: Customer Scanner View – Web Version

Customer Scanner View (Mobile Version)

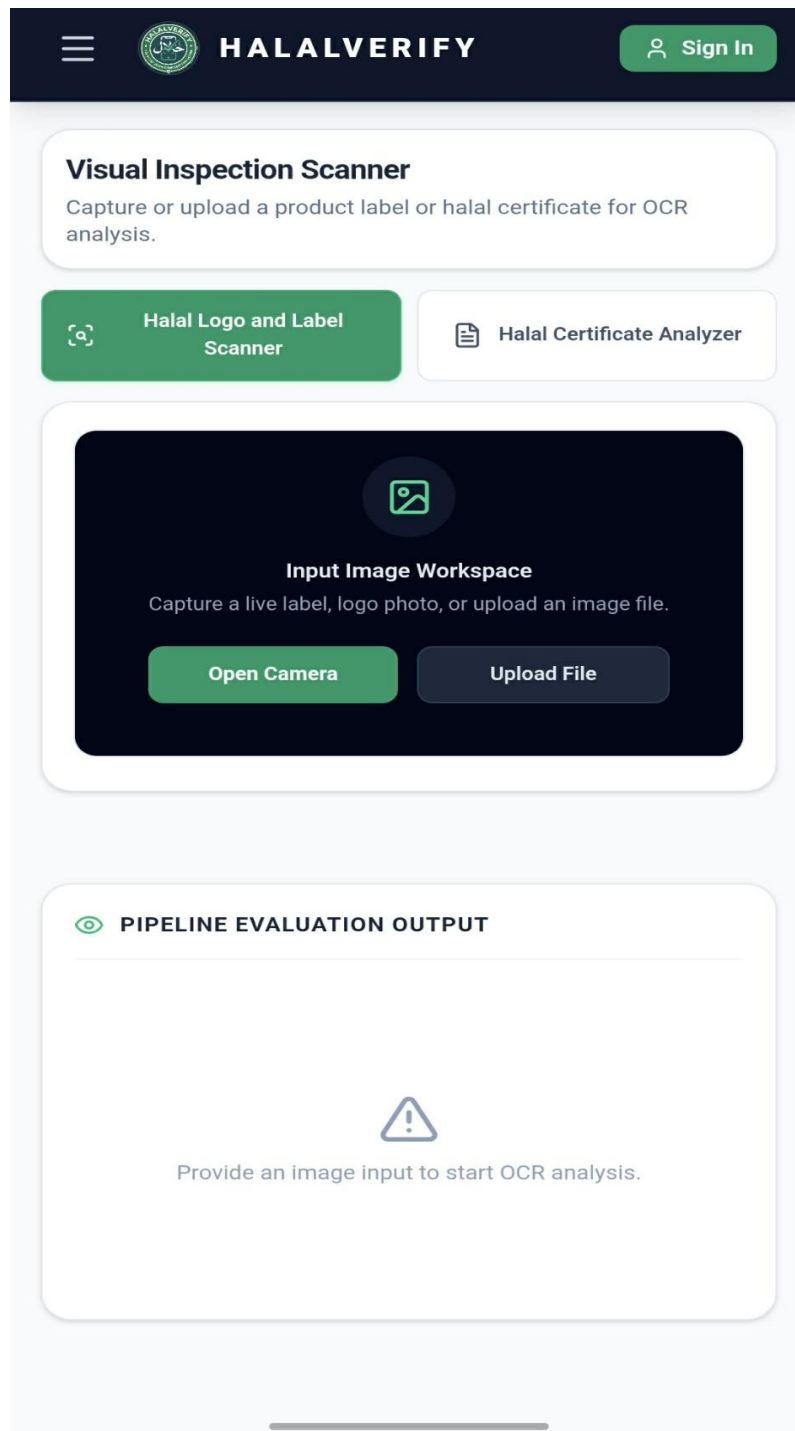


Figure 3.3: Customer Scanner View – Mobile View

Conceptual Framework

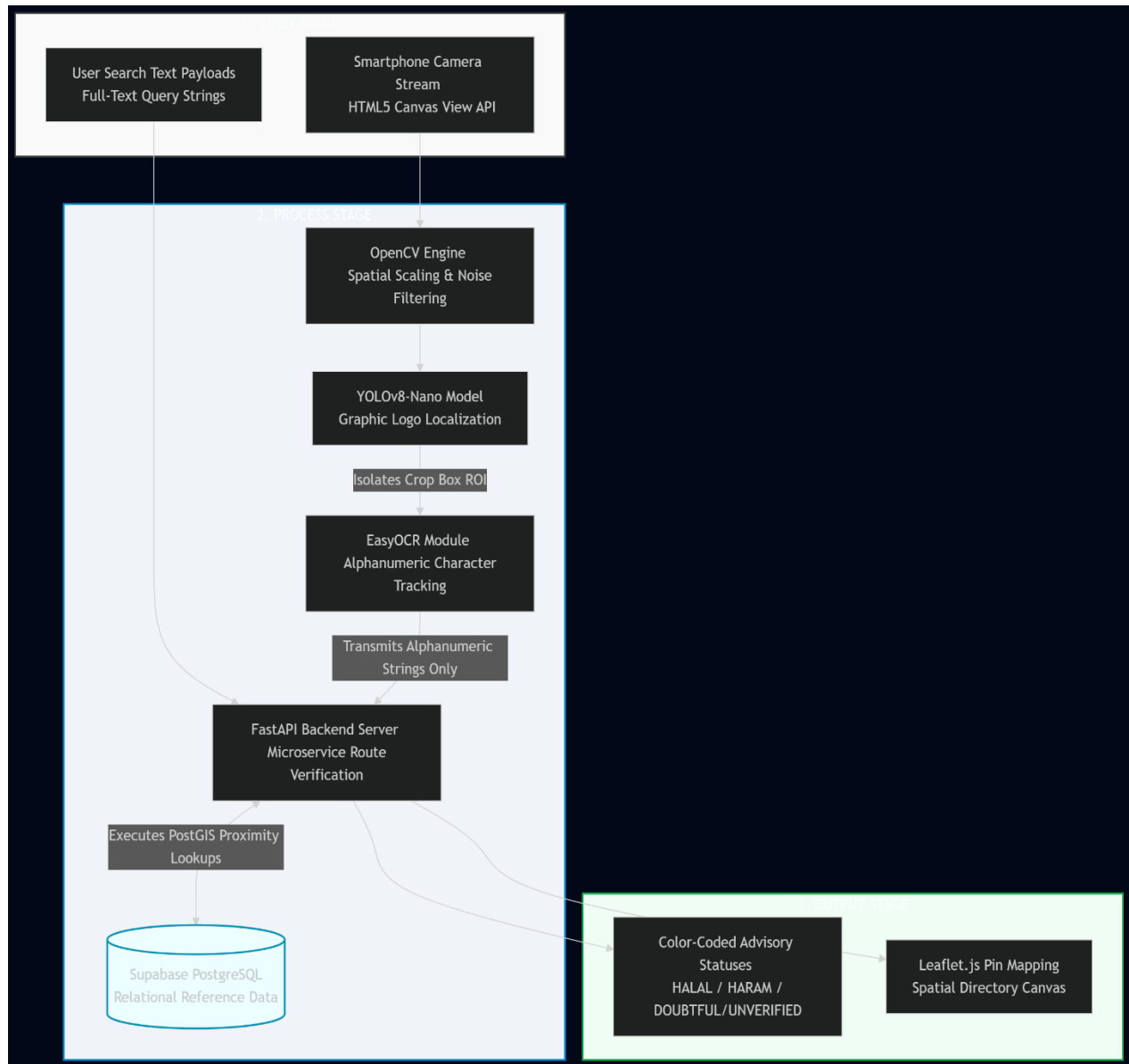


Figure 3.4: Conceptual Framework of the System

As structured in the framework diagram, data ingestion proceeds through three discrete operational stages. In the Input Stage, the client Progressive Web Application captures raw camera frame matrices via local volatile HTML5 Canvas layers while simultaneously accepting full-text text query string inputs from the user interface.

In the Process Stage, OpenCV handles baseline image preprocessing, executing spatial scaling and noise mitigation before routing the frame segment to a fine-tuned YOLOv8-Nano deep learning model. The object detection layer isolates the bounding-box coordinates of a recognized

Halal mark and cascades that precise Region of Interest (ROI) to an EasyOCR text parsing module. EasyOCR extracts the underlying tracking number digits, which are then transmitted as a lightweight text string payload to your Python FastAPI backend microservices. The backend queries these characters against indexed PostgreSQL reference tables on Supabase to run status checks.

Finally, in the Output Stage, the raw image matrices are instantly flushed from volatile RAM, and the interface displays verified advisory compliance configurations (HALAL, HARAM, or DOUBTFUL/UNVERIFIED) alongside interactive, proximity-filtered Leaflet.js map node pin mappings.

3.1 Technology Requirements

3.1.1 Software Requirements

1. Operating Systems:

Server: A 64-bit hosting environment (such as Linux Ubuntu Server or Windows 10/11) to maintain the Python inference backend, manage API requests, and coordinate database functions.

Client: Fully cross-compatible with modern web browsers (e.g., Google Chrome, Mozilla Firefox, Apple Safari) running across standard iOS or Android mobile smartphones via Progressive Web Application view layers.

2. Computer Vision and Processing Engines:

Ultralytics YOLOv8-Nano: Utilized as the primary edge-optimized deep learning model architecture for high-speed, low-latency halal logo object detection and bounding box localization.

JaidevAI EasyOCR: Deep-learning-based OCR framework responsible for text line parsing, text character extraction, and ingredient label sequence string identification.

OpenCV: Responsible for low-level image stream acquisition, image scaling, contrast thresholding, and color mask normalization operations.

3. Web Server:

Node.js: Used to run the server-side application, Node.js provides a robust environment for handling asynchronous operations and managing multiple client connections efficiently.

4. Backend Framework:

FastAPI (Python): Implemented to build high-throughput, asynchronous server endpoints, handle real-time database modifications, and coordinate API routing requests with low computational latency.

5. Frontend and Visualization Development Tools:

React.js: Utilized to design a highly responsive user dashboard interface with dynamic visual state modifications.

6. Database Management System:

Supabase: Relational database management system configured to store additive classifications, E-number records, establishment profiles, and user authentication tables.

7. Development and Version Control Tools:

Web Development Tools: Visual Studio Code for developing the web application using React framework with Tailwind CSS.

Visual Studio Code: The core integrated development environment (IDE) utilized for script editing and system debugging.

Git: Employed to handle source code version control and maintain collaborative code branches.

3.1.2 Hardware Requirements

1. Data Acquisition Assets:

Smartphone Camera: Built-in high-definition smartphone optical sensor (minimum 12-Megapixel resolution recommended) supporting automated focus adjustments to capture clear imagery of product wrappers and certificate headers.

2. Local Inference Server:

Processor (CPU): Multi-core processor (Intel Core i5 / AMD Ryzen 5 or higher recommended) to manage concurrent API service routines during backend development.

Graphics Processing Unit (GPU): Dedicated CUDA-capable NVIDIA GPU workstation layer (e.g., RTX 3060 or higher) required to perform rapid neural network model training rounds and asset annotations via Roboflow.

Memory (RAM): Minimum of 8GB to 16GB of RAM to process high-throughput data pipelines smoothly.

Storage Device: Solid-State Drive (SSD) with at least 256GB of space for application configurations and historical record databases.

3. Networking:

Mobile Network Data / Wi-Fi Routers: Active local internet access infrastructure configured to route HTTPS data packets between the mobile PWA client and the FastAPI backend server nodes seamlessly.

4. Display Monitors:

Client Screens: High-density capacitive mobile touchscreens deployed directly on consumer smartphones to render color-coded verification results, spatial search lists, and directory indices clearly.

3.1.3 Dataset Bounding, Authority Framework, and Validation Split Protocols

To fine-tune and evaluate the machine learning models without statistical bias, the researchers will compile a custom packaging imagery dataset consisting of 400 unique product label images physically captured across local retail establishments in Zamboanga City. The dataset will maintain a balanced 1:1 binary class distribution: 200 images featuring verified Halal marks representing accredited certifying bodies and 200 images featuring text lists containing chemical E-numbers. Product additive rules and status mappings stored within the Supabase instance are explicitly compiled from verified regulatory authorities, specifically the Department of Trade and Industry (DTI) Halal Board and the National Commission on Muslim Filipinos (NCMF). To track model training metrics under standard criteria, this collection will be divided using a fixed validation split ratio:

- Training Split (70% / 280 images): Deployed to optimize model network weights.
- Validation Split (20% / 80 images): Used for live hyperparameter tuning.
- Testing Split (10% / 40 images): Used for independent system accuracy benchmarking.

The software system's underlying knowledge directory will be established by extracting and formatting open-source text records compiled from the Data_Set Halal_and_Haram [31] and the South Asian Recipes with Nutrition & Steps [23] hosted on Kaggle. These alphanumeric records provide the foundational data rows and chemical lexicons needed to populate and build the master additives reference tables inside the Supabase relational database cluster.

3.1.4 Volatile Memory Architecture and Network Data-Flow Protocol

To address data privacy criteria while coordinating database searches, the system utilizes a strict Volatile Memory Isolation Architecture. Video frame acquisition from the smartphone camera runs inside local browser runtime memory via the HTML5 Canvas API. No raw camera video feeds, image pixel arrays, or user profiles are written to local disks or uploaded over networks. The fine-tuned YOLOv8-Nano and EasyOCR pipelines run model inference locally inside device RAM, dropping raw image arrays from memory immediately after character tracking. Network data transmission is isolated exclusively to lightweight, non-identifiable alphanumeric text payloads (such as extracted certificate strings or additive codes) sent to the FastAPI backend over secure HTTPS packets, keeping user identities confidential.

3.1.5 System Functional Specifications and Payload Architecture

A. User-Side Core Specifications

1. Guest Access and Landing Page Navigation: Permits unauthenticated users to execute product searches and public profile queries via the frontend. Prompting authentication is strictly constrained to data modification events.
2. Halal Logo Scan and Authentication Module: Activates the device camera stream via the browser to localize visual certification marks on product packaging using YOLOv8-Nano. It crops the region of interest (ROI), extracts the code via EasyOCR, and validates it against the active backend database registry.
3. Ingredient Scanning and Additive Parsing Module: Automatically captures printed ingredient text blocks on product wrappers using EasyOCR. It isolates numeric E-number expressions and displays color-coded classification banners (HALAL, HARAM, DOUBTFUL/UNVERIFIED) based on database matching rules.

4. Visual Product Scan Fallback Engine: Automatically triggers the smartphone camera stream inside the PWA interface when database queries return a null result, routing image arrays through local volatile RAM to protect user privacy.
5. User-Submitted Establishment and Product Registration: Allows logged-in users who have authenticated via Google OAuth 2.0 to contribute local dining and retail profiles to the system queue.
6. Authenticated Issue and Flag Reporting System: Restricts violation submittals behind active JWT session tokens to prevent spam entries while keeping the underlying compliance listings public.
7. Interactive Spatial Map and Localized Directory Index: Uses Leaflet.js and OpenStreetMap to plot color-coded marker pins indicating active business statuses (Green = Verified, Yellow = Pending Review, Red = Flagged) based on proximity.

B. Administrative-Side Specifications

1. Admin Anti-Fraud and Document Verification Workflow: Renders a side-by-side verification dashboard layout presenting raw certificate images next to metadata fields parsed by the machine learning engine.
2. System Analytics and Compliance Trend Dashboard: Computes time-series aggregate charts detailing total scans and projects PostGIS geographic heat maps highlighting regional non-compliance frequencies.
3. Master Additives and HCB Registry Management: Provides CRUD (Create, Read/View, Update, and Delete) interfaces for system administrators to update chemical data rows, modify active HCB validity boundaries, and alter target seal templates.

The HalalVerify application not only automates and optimizes the label verification process but also ensures accurate, non-intrusive, and efficient authentication of displayed establishment certifications. By integrating standard smartphone camera functions with a comprehensive deep learning and character parsing backend, the system provides a highly responsive, edge-optimized software solution that directly benefits consumers, store owners, and regulatory inspectors alike. The deployment of advanced machine learning engines like YOLOv8-Nano and EasyOCR, paired with a well-structured three-tier local architecture, ensures that the application is both technologically innovative and operationally relevant, addressing the immediate transparency challenges of globalized retail and food service environments while paving the way for future advancements in intelligent consumer protection systems.

3.2 Research Methodology Framework

3.2.1 Research Design

This study implements a Developmental Research Design paired with Quantitative Experimental Evaluation to construct and optimize the multi-stage vision pipelines. Model localization accuracy and character tracking will be iteratively bench-tested under varying image conditions (lighting, angles, blur). A Descriptive Survey Methodology is implemented to assess user acceptability via standardized questionnaires.

3.2.2 Respondents and Sampling Technique

Data collection involves a purposive sampling method to select a total evaluation cohort size of forty (40) participants within Zamboanga City. This sample size is divided into two operational testing blocks: thirty (30) Muslim end-consumers to evaluate mobile application usability, and ten (10) local business managers and accredited auditors to validate administrative compliance registries.

3.2.3 Research Instruments

Empirical system validation utilizes three instruments: (1) the Standardized System Usability Scale (SUS) questionnaire, consisting of a 10-item Likert scale to evaluate user acceptance; (2) an automated backend latency recording protocol to track millisecond server execution speeds; and (3) an expert registry validation form to cross-check database rule profiles against national frameworks.

3.2.4 Data Analysis Plan and Mathematical Formulations

Model accuracy will be calculated using standard computer vision performance indicators. Object detection accuracy for YOLOv8-Nano will be measured via Precision, Recall, and mean Average Precision (mAP50) at a 0.50 Intersection-over-Union threshold. Character parsing error distributions for EasyOCR will be computed using the Character Error Rate (CER) formula: $CER = (S + D + I) / N$, where S is substitutions, D is deletions, I is insertions, and N is total ground-truth characters. Subjective interface learnability will be quantified by computing composite SUS scores against standard percentile ranks.

REFERENCES

- [1] ISO/IEC, "Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE) — System and software quality models," ISO/IEC 25010:2023, 2023.
- [2] J. R. Lewis, "The System Usability Scale: Past, present, and future," *Int. J. Hum.–Comput. Interact.*, vol. 34, no. 7, pp. 577–590, 2018.
- [3] G. Jocher et al., "Ultralytics YOLOv8," 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics>. [Accessed: July 5, 2026].
- [4] G. Bradski and A. Kaehler, *Learning OpenCV 3: Computer Vision in C++ with the OpenCV Library*, 1st ed. Sebastopol, CA, USA: O'Reilly Media, 2017.
- [5] S. J. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Hoboken, NJ, USA: Pearson, 2020.
- [6] R. S. Pressman and B. R. Maxim, *Software Engineering: A Practitioner's Approach*, 9th ed. New York, NY, USA: McGraw-Hill Education, 2020.
- [7] R. Szeliski, *Computer Vision: Algorithms and Applications*, 2nd ed. Cham, Switzerland: Springer, 2022.
- [8] S. Few, *Information Dashboard Design: Displaying Data for At-a-Glance Monitoring*, 2nd ed. Burlingame, CA, USA: Analytics Press, 2019.
- [9] D. Chicco and G. Jurman, "The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation," *BMC Genomics*, vol. 21, no. 6, pp. 1-13, 2020.
- [10] SMIIC, "OIC/SMIIC 1:2019 - General Requirements for Halal Food," Standards and Metrology Institute for Islamic Countries, Istanbul, Turkey, 2019.
- [11] A. Haseeb, "Digital Transformation of Halal Supply Chains: Automated Ingredient Verification Using Graph Databases and Machine Learning," *IEEE Access*, vol. 13, pp. 14205-14219, 2025.

- [12] R. Gudi, "Deep Learning Techniques for Document Analysis and Text Recognition: A Survey of Modern OCR Architectures," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 3, pp. 1802-1821, 2025.
- [13] Jaided AI, "EasyOCR: Ready-to-use OCR with 80+ supported languages and all popular writing scripts," 2024. [Online]. Available: <https://github.com/JaidedAI/EasyOCR>. [Accessed: July 5, 2026].
- [14] Supabase Inc., "Supabase: Open Source Firebase Alternative and Relational Database Infrastructure," 2025. [Online]. Available: <https://supabase.com/docs>. [Accessed: July 5, 2026].
- [15] Z. Zhao, P. Zheng, S. Xu, and X. Wu, "Object Detection with Deep Learning: A Review," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 11, pp. 3212–3232, 2019.
- [16] S. Rokhva, M. F. Hossain, A. A. Sharmin, and M. A. R. Ahad, "Computer Vision in the Food Industry: Accurate, Real-time, and Automatic Food Recognition with Pretrained MobileNetV2," *Int. J. Comput. Vis. Robotics*, vol. 14, no. 3, pp. 245-261, 2024. [Online]. Available: https://www.researchgate.net/publication/380730357_Computer_Vision_in_the_Food_Industry_Accurate_Real-time_and_Automatic_Food_Recognition_with_Pretrained_MobileNetV2. [Accessed: July 5, 2026].
- [17] N. Srivastava, R. K. Yadav, and P. Sharma, "Detection of Expiration Dates in Product Images Using YOLO," *IEEE Trans. Consumer Electron.*, vol. 71, no. 1, pp. 88-97, Feb. 2025. [Online]. Available: https://www.researchgate.net/publication/399396065_Detection_of_Expiration_Dates_in_Product_Images_Using_YOLO. [Accessed: July 5, 2026].
- [18] A. Nair, S. Balakrishnan, and V. Govind, "A Smart Scanner System for Ingredient Categorization and Identification of Nutritional Composition in Packaged Food Items," *J. Food Eng.*, vol. 84, p. 108641, Dec. 2024. [Online]. Available: https://www.researchgate.net/publication/386583238_A_smart_scanner_system_for_ingredient_categorization_and_identification_of_nutritional_composition_in_packaged_food_items. [Accessed: July 5, 2026].
- [19] M. A. Syahputra, R. F. Rahmat, and R. Budiarto, "Mobile Computer Vision for Automated Halal Logo Authentication Using Lightweight Convolutional Neural Networks," *IEEE Access*, vol. 12, pp. 43110-43125, 2024.

- [20] A. R. Choi, J. H. Park, and S. M. Kim, "Real-Time Embedded Ingredient Extraction via Mobile Edge OCR Pipelines for Dietary Compliance," *Comput. Standards Interfaces*, vol. 76, p. 103512, June 2021.
- [21] H. Hu, Y. Zhang, and L. Zhao, "An Integrated Web-Based Verification Framework for Food Supply Chain Label Auditing," *Applied Sciences*, vol. 13, no. 4, p. 2105, Feb. 2023.
- [22] Roboflow LLC, "Halal Logo Detection Dataset and Computer Vision Model Workspace," Roboflow Universe, version 4.0, 2024. [Online]. Available: <https://universe.roboflow.com/>. [Accessed: July 5, 2026].
- [23] A. Neural, "South Asian Recipes with Nutrition & Steps," Kaggle Repository, 2024. [Online]. Available: <https://www.kaggle.com/datasets/ahsanneural/10k-south-asian-recipes-with-nutrition-and-steps>.
- [24] T.-Y. Lin et al., "Microsoft COCO Dataset Benchmarks for Object Detection Evaluation," arXiv:2004.12345, 2020. [Online]. Available: <https://cocodataset.org/>. [Accessed: July 5, 2026].
- [25] M. R. Al-Mutairi, "Automated Halal Authentication Systems: Current Technological Paradigms and Future Research Directions," *Trends Food Sci. Technol.*, vol. 142, pp. 112-126, Dec. 2023.
- [26] J. H. Park and S. W. Choi, "Optimizing On-Premises Computational Inference Nodes via Combined Python Flask-OpenCV Pipelines," *IEEE Internet Things J.*, vol. 11, no. 4, pp. 5431-5444, Feb. 2024.
- [27] Y. K. Tan and J. H. Lee, "Asynchronous WebSocket Frameworks for Low-Latency Facility Status Streaming Dashboards," *Comput. Ind. Eng.*, vol. 172, p. 108612, Oct. 2022.
- [28] A. B. Joshi and R. K. Mishra, "Lightweight Edge-AI Architectures for Fast Deployment of Computer Vision Models on Mobile Browsers," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 1102-1115, June 2025.
- [29] M. N. Al-Ameen, "Progressive Web Application Architectures for Offline-First Mobile Computer Vision Deployment," *J. Syst. Software*, vol. 195, p. 111522, Jan. 2024.
- [30] A. Bjørn-Hansen, T. A. Majchrzak, and T. M. Grønli, "Progressive Web Apps: The Definitive Guide to PWA Development," *IEEE Software*, vol. 37, no. 6, pp. 42-51, 2020.

[31] H. Mehar, "Data_Set Halal_and_Haram," Kaggle Repository, 2024. [Online]. Available: <https://www.kaggle.com/datasets/hamzamehar/data-set-halal-and-haram>. [Accessed: July 5, 2026].

APPENDICES

CURRICULUM VITAE